

Lyons Health and Safety Advisory Panel

Meeting of March 22, 1993

PARTICIPANTS: Greg Wilkinson, Rob Terry, John Martin new member and ad hoc rep of BREATHE, Milt Lammering, Bob Trujillo, Anne Lovell for Joni Teter, Jan Johnson, John Lohr.

ITEMS DISCUSSED:

New members were introduced. John Martin is a professor of Mathematics at CSU and will serve as an ad hoc representative of BREATHE although he is not a formal member. Jane Shellenberger has agreed with this approach.

Anne Lovell was in attendance for Joni Teter.

The meeting was consumed after this point with discussions of analytical labs, the sampling matrix, and a sampling firm.

SAMPLES

The sampling matrix was finalized as shown on the attached sheet. The key changes were:

1. Attempt to obtain core samples of the CKD in the quarry in order to obtain a history on this material. It is uncertain whether it is feasible to obtain core samples. If not then 2 or 3 samples from surface piles will be used.
2. The raw material matrix was simplified by eliminating the high cal material from the Larimer County quarry as not being significant new information.
3. It was specified that the high cal material from the Lyons quarry should be 2 samples at different depths from the same bore hole.
4. It was specified that the background samples should be taken from the following locations:
 - a. 1 sample from the SWPC plant site
 - b. 1 sample from the Dow Flats area since it is undisturbed and upwind of the plant.
 - c. 1 sample from an area between the plant and Hygiene since this will be downwind.
 - d. 1 sample from Livermore since this is away from any potential plant influence and has a similar setting as Lyons. This sample will come from the CO Department of Health from material

collected in the last year.

5. It was agreed that the urine samples from employees would not be collected at this time. It would be more appropriate to see the results of the other materials and if significant levels of Plutonium are detected, then it would be appropriate to test the levels in humans.

LABORATORIES

After discussion of the test methodology, accuracy, lead times, and the merits of three (3) separate testing labs, the following was agreed:

Acculabs was selected to be "Lab 1" as identified on the sample matrix.

- Capable of counting to the required accuracy
- Very experienced in Plutonium testing
- Good reputation for reliability and accuracy
- Reasonable price
- 8 to 10 week lead and test time

Colorado State University was selected as "Lab 2" as identified on the sample matrix.

- Able to reach or come close to required accuracy
- Experienced in Plutonium testing
- Good reputation for reliability and accuracy
- More expensive but competitive
- 60 day lead and test time

A third laboratory was thrown out because of limited experience in plutonium measurement.

OBTAINING SAMPLES

There were two engineering firms discussed for actually pulling the samples for testing: SHB AGRA and Sheperd Miller.

SHB AGRA has prior experience but has a consulting relationship with SWPC currently. This relationship eliminated them to avoid any conflict of interest or bias.

Sheperd Miller was recommended by Jan Johnson as having prior experience and capable of doing the job requested. As a matter of elimination it was agreed that they would be the sampling firm.

It was agreed that the sampling date, time and location would be made public and that representatives of interested groups would be invited. The only concern is having too large a

crowd which could be a safety and logistics issue. If the response is strong a procedure for selecting representatives will be devised.

In addition, Sheperd Miller will be required to insure a chain of custody procedure be established for all samples.

John Lohr will contact Sheperd Miller to set up procedures and dates for sampling.

WRAP-UP

It was tentatively agreed that the next meeting would be the day of the sampling. If other issues arise, the Panel will be convened.

Lyons Health and Safety Advisory Panel

Plutonium Sampling Recommended:

	EPA		LAB	
	1	2	1	2
Confirm EPA Findings				
EPA split sample of CKD	1		1	1
EPA split sample of CKD as managed				
EPA split sample of Clinker	1		1	1
Replicate EPA Sampling with new Samples				
Sample of CKD from Alkalai Baghouse			1	1
Sample of CKD in Quarry (Core Sample)			3	3
Sample of Clinker			1	1
Evaluate Raw Materials				
Weathered 4th Ridge (Surface Material)		3	3	
Lyons Silica (Surface Exposure)		2	2	
High Cal from Lyons, 60 ft below surface		2	2	
Used motor oil from old pipeline		1	1	1
Establish Controls/Background				
Soil Samples in local area			4	4
Total Samples Recommended	2	8	19	12

COST:

Cost per Analytical @ \$500 EACH + Sampling cost

Cost of Additional Analytical Tests Approx \$15,500

LABS:

EPA 1 is the first set of samples tested @ NAREL in '92

EPA 2 are the samples of raw materials agreed to by EPA
to be run at NAREL in spring of '93

LAB 1 is one independent/private lab; Acculabs

LAB 2 is a second independent/private lab; CSU Lab

BACKGROUND SAMPLES:

1 @ Plant Site

1 between Plant and Hygiene (Downwind)

1 @ Dow Flats (Upwind)

1 @ Livermore (Removed from area but similar terrain)

SAMPLING:

To be done by Shepherd Miller, Inc.

Will be publicly announced and witnessed

Date to be set

Revised on 3/22/93

Summary

Lyons Health and Safety Advisory Panel and Southwestern Portland Cement

- Thank Officials and Community Representatives
Willingness to participate in LHSAP
Patience and hard work to date
Community and official participation insures
objectivity and thoroughness
- Southwestern Portland Cement
Surprised and confused about Plutonium findings
Committed to answering community concerns/questions
Will continue until questions are answered
Supporting and participating in LHSAP
Funding for necessary testing
Proactive participation from the start
- County Commissioner Support and Approval
Continue the work begun
Support LHSAP until the report is issued
Offer suggestions on how to improve the process

Findings and Plans

Lyons Health and Safety Advisory Panel
and Southwestern Portland Cement

- Findings to Date

1. Trace levels not unlike background levels
2. Very small concentrations at or below detection levels
Similar concentrations in control or blank sample
3. Presence of Plut 238 is strange
Would expect Plut 239, fallout residual
Ratio of 238 to 239 is reversed
4. Possible case of mistaken identity?
Dr. Shawki Ibrahim of CSU
Letter to editor
5. Minute concentrations: "Below health risk screening criteria"

- Plans

Complete sampling and testing of:
Raw Materials, Products, CKD, Employees
Multiple samples and dual labs

bcc6

March 9, 1993

Boulder Co Commissioners Meeting

Introduction and Objectives

Lyons Health and Safety Advisory Panel and Southwestern Portland Cement

- John Lohr
Plant Manager at SWPC in Lyons
- Cement Process Overview
- Background and History of Issue
- Actions and Plans by Lyons Health & Advisory Panel

Cement Process Overview

Lyons Health and Safety Advisory Panel
and Southwestern Portland Cement

- Quarry 800,000 tons annually
-shale and limestone
- Rock is crushed and ground to -200 mesh
- Fed into a Rotary Kiln @ 100 tons/hr
14 ft dia x 245 feet
30 to 60 min @ 2700 F.
Clinker is created, cooled & stored
Clinker is ground into Portland Cement
- Kiln is fueled by:
Used motor oil or #6 fuel oil; ended in 9/90
Tires, up to 2.5 tons per hour
Coal, up to 11 tons per hour, 1/3 of fuel
Natural gas, 2/3 of fuel

History of the Issue

Lyons Health and Safety Advisory Panel
and Southwestern Portland Cement

- Cement Industry
Heavy industry
Waste fuels in some kilns
EPA evaluation of process, products & wastes
- EPA Characterizing CKD
Dust collected in baghouses
Raw and partially clinkered material
- Cement Industry Volunteers
Long Questionnaire fall of '91
Sampling of 15 kilns
Lyons identified as a backup
2 hour notice of 3/27/92 sampling

History of the Issue

Lyons Health and Safety Advisory Panel
and Southwestern Portland Cement

- Lab Results Released in Late November
- Concerns Arose in December
 - Manmade substance
 - Were concentrations dangerous?
 - How did it surface in clinker?
 - Allegations arose about nuclear waste disposal at Lyons
- SWPC Begins an Investigation
 - Hire Dr. Greg Wilkinson, Phd, Epidemiologist
 - Consultations with state, local, EPA officials
 - Designed investigative process

bcc4

March 9, 1993

Boulder Co Commissioners Meeting

History of the Issue

Lyons Health and Safety Advisory Panel and Southwestern Portland Cement

- Lyons Health and Safety Advisory Panel
Citizens: Joni Teter, Dan Murphy, Ken Cinnamon
Officials: Robert Terry, Cliff Myers, Lee Phipps
Milt Lammering
Experts: Janet Johnson and Greg Wilkinson
Company: Bob Trujillo, Gerald Anderson, John Lohr
BREATHE: Invited to join prior to Mtg #2
- Objectives of LHSAP
Determine concentrations of Plutonium in plant materials
Compare SWPC materials to concentrations in community
Determine likely sources of Plutonium
Determine any safety of health risks to EE's or community
Inform public and local officials, summer, 1993.